

HumidClimatologyEngine

Complete Scientific Reference, Methodology, Formula Compendium, Validation Guide, and User Manual

Project purpose: reproducible, memory-aware, production-oriented daily moisture climatology from ERA5-Land temperature, dew-point temperature, and surface pressure.

Document language: English Project release inspected: 0.6.1 / schema 6.0 Climatology period configured in source: 1981–2020

Primary outputs: relative humidity (RH), vapor pressure (e), mixing ratio (r), and specific humidity (q), with mean, sample standard deviation, bias-corrected skewness, and Fisher excess kurtosis for each climatological day and grid cell.

Layer	Implementation
Data	ERA5-Land official post-processed daily statistics / hourly fallback
Calendar	366-slot climatology; Feb-28 + Feb-29 pooled into DOY 60; DOY 59 reserved
State model	Joint multivariate normal in (T, Td, ln P)
Numerics	Welford/Pébay accumulation, covariance merging, PSD-aware Cholesky
Propagation	Streaming Monte Carlo; configurable sample count, cell chunks, sample batches
Persistence	Annual/daily checkpoints, SHA-256 verification, configuration/script hashes
Output	Compressed NetCDF4 climatology + diagnostic NetCDF4
Testing	Calendar, numerical statistics, physics ground truth, synthetic MVN, restart/consistency tests

Scientific status: This guide documents the implementation found in the supplied archive. It distinguishes implemented behavior from recommended extensions and does not silently substitute a different scientific method.

Prepared as a project-level reference manual

Based on the supplied HumidClimatologyEngine source archive and cross-checked against primary/official scientific references.

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1. Executive Summary

HumidClimatologyEngine is a scientific software system for converting long ERA5-Land daily archives into a compact climatological description of atmospheric moisture. The central design choice is to avoid treating humidity variables as independent empirical fields. Instead, the engine models a physically meaningful joint state consisting of 2-m air temperature, 2-m dew-point temperature, and the logarithm of surface pressure, then propagates uncertainty and nonlinear thermodynamic relationships through Monte Carlo sampling.

The supplied implementation is explicitly designed around reproducibility and bounded memory. Historical observations are accumulated with online/mergeable statistics, annual states can be checkpointed independently, and the final climatology is generated through streaming NetCDF writes rather than materializing the full Monte Carlo tensor in memory.

The model has five conceptual layers: (1) acquisition and provenance, (2) calendar normalization and paired-valid data screening, (3) statistical estimation of the joint state, (4) nonlinear physical transformation to moisture variables, and (5) distributional summarization plus diagnostics. This separation is important because it makes failures attributable: a problem can be located in data quality, statistics, physical conversion, Monte Carlo validity, or serialization rather than being hidden inside a single monolithic calculation.

The repository configuration sets the main production period to 1981–2020, 366 climatological slots, 5,000 Monte Carlo samples, a 1,024-cell spatial chunk, a 256-sample batch, a conservative two-worker default, and a fixed random seed. These values are implementation defaults, not universal scientific constants; sensitivity analysis should be used when the method is applied to a new domain or research question.

What the engine produces

- For each climatological day and grid cell: RH, vapor pressure, mixing ratio, and specific humidity.
- For each transformed moisture variable: mean, sample standard deviation, bias-corrected skewness, and Fisher excess kurtosis.
- Diagnostics: supersaturation fraction, invalid e/P fraction, invalid covariance fraction, minimum covariance/correlation eigenvalue, valid Monte Carlo sample count, correlations among T, Td, and log(P), and valid observation count.
- NetCDF4 files containing coordinate/calendar metadata, scientific attributes, configuration hash, script SHA-256, schema version, and creation timestamp.

Core scientific idea

$$X = (T, Td, \ln P)^T \sim N(\mu, \Sigma)$$

The engine estimates the empirical mean vector and covariance structure of X for each climatological day and grid cell. Sampling from this joint model preserves the observed linear dependence among temperature, dew point, and pressure at the modeled scale. Each sampled state is then transformed through saturation vapor pressure and moisture equations.

$$e = e_s(Td), \text{ RH} = 100 e_s(Td) / e_s(T), r = 0.622 e / (P - e), q = r / (1 + r)$$

Because the transformations are nonlinear, the climatological moments of RH, e , r , and q are not obtained by simply transforming the means of T , Td , and P . Monte Carlo propagation is therefore a central part of the method.

2. System Architecture and Processing Pipeline

The repository is organized as a small research software package rather than a single script. The principal execution wrapper imports the production engine from `src/moisture_climatology_v6.py`. Acquisition scripts provide an official CDS route and an hourly fallback. Documentation, examples, tests, configuration, and packaging metadata accompany the engine.

2.1 End-to-end pipeline

- 1 Acquire ERA5-Land inputs and record their provenance.
- 2 Audit the downloaded files, variables, units, coordinates, dates, missing values, and time convention.
- 3 Normalize each observation into the project's climatological day-of-year mapping.
- 4 Apply paired-valid filtering: T , T_d , and P must all be finite and pressure must be positive.
- 5 Accumulate annual statistics for T , T_d , and $\ln(P)$ using Welford-style updates and cross-products.
- 6 Persist annual checkpoints with integrity metadata.
- 7 Merge annual states using parallel/mergeable formulas.
- 8 Convert second-order states to standard deviations, covariances, and correlations.
- 9 Construct the daily three-variable dependence model.
- 10 Check positive-semidefiniteness and perform controlled Cholesky factorization.
- 11 Generate Monte Carlo samples in cell chunks and sample batches.
- 12 Transform every valid realization into RH, vapor pressure, mixing ratio, and specific humidity.
- 13 Accumulate transformed-variable moments using Pébay-style higher-moment machinery.
- 14 Stream the climatology and diagnostics into NetCDF4.
- 15 Run final physical, statistical, convergence, and checkpoint integrity validation.
- 16 Archive configuration, hashes, diagnostics, and provenance with the final products.

2.2 Data flow

```
ERA5-Land daily statistics
|
+--> T2m
+--> Td2m
+--> surface pressure
|
v
Calendar normalization + paired-valid mask
|
v
Annual Welford/Pébay states
|
v
Annual checkpoint files
|
v
Parallel state merge
|
v
mu, std, covariance, correlation of (T, Td, lnP)
|
v
PSD / Cholesky validation
|
v
Streaming multivariate Monte Carlo
|
v
Phase-aware saturation vapor pressure
|
+--> RH
+--> e
+--> r
+--> q
|
v
Higher moments + diagnostics
|
+--> moisture_climatology_1981_2020.nc
+--> moisture_climatology_diagnostics_1981_2020.nc
```

2.3 Separation of concerns

Concern	Responsible layer	Primary failure signal
Data acquisition	CDS downloader scripts	Missing/incorrect files or retrieval errors
Data validity	Input/calendar accumulation	Low paired-valid counts
Numerical statistics	Welford/Pébay states	Mismatch against NumPy/SciPy tests
Dependence model	Covariance/correlation builder	Invalid or non-PSD matrices
Physical conversion	saturation_vapor_pressure / derive_moisture	Bounds, $e \geq P$, supersaturation
Monte Carlo	Daily streaming sampler	Low valid sample counts / non-convergence
Persistence	NetCDF finalizer	Missing variables / checksum mismatch

3. Repository Map and Software Components

The supplied archive was inspected directly. The following map reflects the files present in the archive rather than an assumed generic Python project.

Path	Role
README.md	Project overview, scientific rationale, formulas, workflow, validation, limitations, and usage.
src/moisture_climatology_v6.py	Main production implementation; statistics, physics, Monte Carlo, checkpoints, tests, and logging.
moisture_climatology.py	Thin execution wrapper that imports main from the v6 source.
scripts/download_era5land_daily_statistics.py	Official CDS downloader for ERA5-Land post-processed daily statistics.
scripts/download_era5land_hourly.py	Hourly ERA5-Land fallback downloader.
docs/methodology.md	Condensed scientific methodology.
examples/production_run.md	Recommended production execution sequence.
examples/download_2021_example.md	Official CDS example and legacy-data provenance guidance.
examples/HumidClimatologyEngine_tutorial.ipynb	Teaching/tutorial notebook.
tests/test_repository.py	Repository-level tests for leap-day mapping, physics, and logging consistency.
environment.yml	Conda environment definition.
requirements.txt	Python dependency specification.
pyproject.toml	Package metadata and optional test/notebook dependencies.
CITATION.cff	Software citation metadata.
CHANGELOG.md	Release history and v0.6.1 bug-fix notes.
CONTRIBUTING.md / SECURITY.md / LICENSE	Contribution, credential-security, and licensing guidance.

3.1 Package metadata

The inspected pyproject declares package version 0.6.1, requires Python ≥ 3.11 , and lists NumPy, SciPy, xarray, netCDF4, tqdm, psutil, cdsapi, and PyYAML as core dependencies. Test and notebook functionality is exposed through optional dependency groups.

3.2 Reproducibility metadata

- `SCHEMA_VERSION = 6.0` and `CHECKPOINT_VERSION = 6.0` in the production source.
- A fixed `RANDOM_SEED` is configured for Monte Carlo reproducibility.
- A configuration hash and script SHA-256 are propagated into output metadata.
- Annual and daily checkpoints are separated, supporting restart and partial reruns.

3.3 Important distinction

The source contains a production wrapper named `moisture_climatology.py`, while the main scientific implementation lives under `src/`. When modifying the project, the source of truth for the scientific algorithm is the v6 module; the wrapper should remain intentionally thin.

4. Input Data: ERA5-Land and Acquisition

The preferred acquisition route in the supplied project is the Copernicus Climate Data Store (CDS) post-processed ERA5-Land daily statistics product. The official CDS catalogue describes this product as daily statistics generated from ERA5-Land hourly data during retrieval. It provides daily mean, minimum, and maximum options, selectable sub-daily sampling frequency, and a configurable UTC offset for local-day aggregation. The catalogue currently reports global $0.1^\circ \times 0.1^\circ$ coverage, a temporal range beginning in January 1950, and NetCDF-in-ZIP delivery.

`□cite□turn0search0□turn0search2□`

4.1 Required physical variables

Concept	Default source variable in project	Expected unit	Internal representation
2-m air temperature	average_t2m	K	°C
2-m dew-point temperature	average_d2m	K	°C
Surface pressure	average_sp	Pa	hPa, then $\ln(\text{hPa})$ for statistics

4.2 Unit conversions

$$T_C = T_K - 273.15$$

$$Td_C = Td_K - 273.15$$

$$P_{\text{hPa}} = P_{\text{Pa}} / 100$$

$$L = \ln(P_{\text{hPa}})$$

Pressure is modeled in logarithmic space because pressure is strictly positive. A Gaussian distribution in raw pressure can theoretically produce negative draws; a Gaussian distribution in log-pressure guarantees positive reconstructed pressure after exponentiation.

4.3 Official daily-statistics route

```
python scripts/download_era5land_daily_statistics.py \  
--years 2021 \  
--variables 2m_temperature 2m_dewpoint_temperature surface_pressure \  
--statistics daily_mean daily_minimum daily_maximum \  
--out-dir ./data/raw
```

The project explicitly recommends comparing historical manually aggregated data with the official CDS-derived product before substitution, including daily definitions, time-zone convention, sampling frequency, missing values, and leap-day handling.

4.4 ERA5-Land scientific provenance

ERA5-Land is a replay of the land component of ERA5 and uses atmospheric forcing from ERA5. The CDS documentation emphasizes that the dataset is a reanalysis/model estimate rather than direct observations at every grid cell, and that uncertainty generally increases farther back in time as observational constraints become sparser. `□cite□turn0search2□turn0search14□`

4.5 Acquisition best practice

- Freeze the exact retrieval request: dataset, variables, years, months, daily statistic, frequency, time zone, and area.
- Store the retrieval request alongside the downloaded archive.
- Do not commit CDS credentials or API tokens.

- Record file names, sizes, checksums, and retrieval timestamps.
- Never silently mix UTC-day data with local-day data.
- If legacy manual daily data are retained, label them as a distinct processing lineage.

5. Calendar Model and Leap-Day Convention

The engine deliberately uses 366 climatological slots. Its special rule is: February 28 and February 29 are pooled into climatological DOY 60; DOY 59 is reserved; March 1 maps to DOY 61. This is a custom scientific convention and should be treated as part of the model definition, not as an incidental indexing detail.

5.1 Mapping

Gregorian date	Leap year DOY	Non-leap year DOY	Climatological slot
Feb 28	59	59	60
Feb 29	60	—	60
Mar 1	61	60	61
Mar 2	62	61	62

5.2 Why explicit handling matters

- A one-day shift after February can corrupt every seasonal statistic from March onward.
- Pooling February 28 and 29 changes the sampling distribution of that climatological slot.
- The reserved DOY 59 must remain semantically documented in downstream software.
- Any comparison with a conventional 365-day climatology requires an explicit remapping.

5.3 Implemented test

```
assert get_clim_doy(59, 1984) == 60
assert get_clim_doy(60, 1984) == 60
assert get_clim_doy(61, 1984) == 61
assert get_clim_doy(59, 1985) == 60
assert get_clim_doy(60, 1985) == 61
```

5.4 Recommended alternative analyses

- Compare pooled Feb-28/29 against a 365-day climatology with Feb-29 removed.
- Compare pooled handling against separate leap-day modeling when enough years are available.
- Quantify sensitivity of winter-to-spring transitions to the calendar convention.

6. Statistical Core: Paired-Valid Joint State

For every climatological day and grid cell, the engine considers a sample valid only when temperature, dew point, and pressure are all finite and pressure is positive. This paired-valid rule is essential: independently filtering the three variables would destroy their joint covariance structure.

$$X_i = [T_i, Td_i, \ln(P_i)]^T$$

For n valid observations, the empirical mean vector is

$$\mu = (1/n) \sum X_i$$

and the sample covariance matrix is

$$\Sigma = (1/(n-1)) \sum (X_i - \mu)(X_i - \mu)^T$$

The three variances occupy the diagonal and the three unique cross-covariances occupy the off-diagonal entries.

$$\Sigma = \begin{bmatrix} s_T^2 & \text{cov}(T, Td) & \text{cov}(T, L) \\ \text{cov}(T, Td) & s_{Td}^2 & \text{cov}(Td, L) \\ \text{cov}(T, L) & \text{cov}(Td, L) & s_L^2 \end{bmatrix}$$

6.1 Correlations

$$\rho_{XY} = \text{cov}(X, Y) / (s_X s_Y)$$

The implementation clips correlation estimates to the interval $[-0.999999, 0.999999]$ before covariance reconstruction/Cholesky processing. This protects against numerical endpoint problems while preserving the sign and near-perfect dependence.

6.2 Minimum observation requirement

The configured `MIN_OBS` is 3. However, standard deviations and covariances require at least two observations, and higher-order distributional summaries require more. The engine therefore carries explicit validity states rather than manufacturing statistics from insufficient samples.

6.3 Why paired-valid is scientifically important

Suppose T is available for 100 observations but Td is available for only 80, and pressure is available for 90. The joint model cannot safely combine separate marginal sample sizes because covariance requires paired values. The engine instead uses the intersection of valid observations so every row of X corresponds to a real joint atmospheric state.

7. Welford Online Statistics and Parallel Merging

Long climate archives can be too large to hold in memory. The engine therefore uses online statistics. Welford's 1962 method provides a numerically stable update for corrected sums of squares and is the conceptual foundation of the implementation. [cite]turn0search1[turn0search3]

7.1 Scalar Welford update

For a new value x , previous count n , previous mean μ , and previous sum of squared deviations $M2$:

```
n' = n + 1
delta = x - mu
mu' = mu + delta/n'
M2' = M2 + delta (x - mu')
```

After processing n observations, the sample variance is

```
s^2 = M2/(n-1), n >= 2
```

The source applies this update vectorized across valid spatial cells, which gives the memory efficiency of online statistics without forcing a Python loop over every grid point.

7.2 Cross-products and covariance

For paired variables x and y , the cross-deviation accumulator C_{xy} can be updated in a compatible online form. At the end,

```
cov(x,y) = C_xy/(n-1)
```

The implementation stores C_{T_Td} , C_{T_logP} , and C_{Td_logP} alongside the three scalar $M2$ states.

7.3 Parallel merge

Annual checkpoint states can be merged without replaying all raw observations. For state A with $n1$, mean $m1$, $M2_1$ and state B with $n2$, mean $m2$, $M2_2$:

```
n = n1 + n2
delta = m2 - m1
m = m1 + delta (n2/n)
M2 = M2_1 + M2_2 + delta^2 (n1 n2/n)
```

For a covariance cross-product C_{xy} , the same correction becomes

```
C_xy = C_xy,1 + C_xy,2 + (m_x2-m_x1)(m_y2-m_y1)(n1 n2/n)
```

This is what makes annual checkpoints scientifically useful rather than merely operational: each checkpoint is a sufficient mergeable statistical state for the second-order joint model.

7.4 Numerical advantages

- Avoids repeated catastrophic cancellation from naive $\text{sum}(x^2) - n \cdot \text{mean}(x)^2$ formulas.
- Allows year-by-year processing and restart.
- Allows independent validation of annual states before global merging.
- Makes the computation scalable to multi-decade, high-resolution grids.

8. Higher Moments and Pébay-Style Moment Algebra

After nonlinear transformation to RH, vapor pressure, mixing ratio, and specific humidity, the engine retains more than the mean and standard deviation. It computes skewness and Fisher excess kurtosis, which provide information about asymmetry and tail shape.

8.1 Central moment definitions

$$M_k = \sum (x_i - \bar{x})^k$$

$$m_k = M_k/n$$

The second central moment m_2 is related to variance. The third and fourth moments describe asymmetry and tail weight.

8.2 Bias-corrected skewness

The implementation follows the SciPy `bias=False` convention. With n observations and $b_2 = M_2/n$:

$$G_1 = \sqrt{n(n-1)} / (n-2) * m_3 / m_2^{3/2}, n \geq 3$$

8.3 Fisher excess kurtosis

$$b_2 = n M_4 / M_2^2$$

$$G_2 = (n-1) / [(n-2)(n-3)] * [(n+1)b_2 - 3(n-1)], n \geq 4$$

Fisher excess kurtosis is zero for a Gaussian distribution in the population limit. Positive values indicate heavier tails than Gaussian; negative values indicate lighter tails.

8.4 Batch accumulation

The Monte Carlo engine processes samples in batches. Each batch obtains n , mean, M_2 , M_3 , and M_4 along the sample axis, then merges those states. This is a Pébay-style strategy: moments are accumulated in mergeable form rather than storing every realization.

8.5 Interpretation warning

Higher moments can be substantially noisier than means. A stable-looking mean does not imply stable kurtosis. In production studies, convergence should be evaluated separately for each statistic and variable.

9. Thermodynamic Transformation Layer

The physical layer converts sampled T , T_d , and pressure into moisture variables. This is where the joint statistical model becomes scientifically interpretable.

9.1 Saturation vapor pressure over water

$$e_{s,w}(T) = 6.112 \exp[17.67 T/(T + 243.5)]$$

The project implements the Magnus-type water expression above, with T in $^{\circ}\text{C}$ and e_s in hPa.

9.2 Saturation vapor pressure over ice

$$e_{s,i}(T) = 6.112 \exp[22.46 T/(T + 272.62)]$$

For phase awareness, the implementation uses water for $T \geq 0^{\circ}\text{C}$ and ice for $T < 0^{\circ}\text{C}$. A related Magnus formulation based on Sonntag (1990) uses closely related coefficients; for example, 17.62/243.12 over water and 22.46/272.62 over ice. The exact coefficient set matters for reproducibility, so this project documents its own implemented equations rather than treating all Magnus formulations as interchangeable. [\[cite\]turn0search36](#)

9.3 Vapor pressure from dew point

$$e = e_s(T_d)$$

Dew point is the temperature at which the actual vapor pressure reaches saturation. Therefore, applying the saturation formula to T_d provides the water-vapor partial pressure under the adopted approximation.

9.4 Relative humidity

$$\text{RH_raw} = 100 e_s(T_d) / e_s(T)$$

$$\text{RH} = \min(100, \max(0, \text{RH_raw}))$$

The implementation records supersaturation before clipping. This is scientifically preferable to silently clipping first because it preserves a diagnostic of how often the sampled joint model produces $\text{RH_raw} > 100\%$.

9.5 Mixing ratio

$$r = \epsilon e / (P - e), \epsilon = 0.622$$

The implementation requires $e > 0$, $P > 0$, and $e < P$ before computing r . If $e \geq P$, mixing ratio is invalid and the corresponding realization is rejected from the valid transformed sample.

9.6 Specific humidity

$$q = r / (1 + r)$$

The mixing-ratio and specific-humidity definitions are consistent with standard atmospheric thermodynamics. Classical meteorological references define vapor pressure, saturation vapor pressure, mixing ratio, and specific humidity through the same pressure-partitioning relationships. [\[cite\]turn0search37](#)

10. Multivariate Normal Model

For each climatological day and grid cell, the engine approximates the empirical joint distribution by a three-dimensional multivariate normal distribution in transformed state space:

$$X = (T, Td, \ln P)^T \sim N(\mu, \Sigma)$$

10.1 Why a joint model?

- T and Td are physically coupled because dew point is constrained by atmospheric moisture.
- Pressure is correlated with temperature and synoptic regime.
- Derived moisture variables are nonlinear functions of all three state variables.
- Independent marginal sampling would destroy the empirical covariance structure.

10.2 Why log pressure?

Pressure must remain positive. Modeling $\ln P$ allows the Gaussian sampler to operate on an unconstrained real-valued variable while guaranteeing $P = \exp(\ln P) > 0$ after reconstruction.

10.3 Covariance matrix

$$\Sigma = D R D$$

where $D = \text{diag}(s_T, s_{Td}, s_{\ln P})$ and R is the correlation matrix. The source derives correlations from empirical covariance estimates and uses a bounded representation before factorization.

10.4 Gaussian assumption

The Gaussian approximation is a pragmatic, computationally tractable model. It is not a claim that atmospheric state distributions are exactly Gaussian. The project itself identifies the multivariate-normal assumption as a central limitation and recommends sensitivity tests against mixture or copula alternatives.

10.5 When the Gaussian approximation is most vulnerable

- Strongly bimodal synoptic regimes.
- Mountain/coastal cells with mixed air-mass populations.
- Very cold regimes with phase transitions in moisture physics.
- Rare-event statistics where tail dependence matters more than central moments.
- Cells with very small sample sizes.

11. Positive-Semidefinite Covariance and Cholesky Factorization

A valid covariance/correlation matrix must be positive semidefinite. Floating-point accumulation and aggressive correlation clipping can produce matrices that are microscopically negative in one eigenvalue. The implementation therefore performs an explicit eigenvalue check before Cholesky factorization.

11.1 Eigenvalue criterion

$\lambda_{\min}(R) \geq 0$ (ideal mathematical condition)

The configured numerical tolerance allows small negative eigenvalues down to `PSD_REPAIR_TOL` in magnitude to be treated as repairable. Larger violations are considered hard-invalid and are skipped rather than silently forced into a valid matrix.

11.2 Cholesky factor

$$R = L L^T$$

If $z \sim N(0, I)$, then

$$y = \mu + L z$$

has covariance LL^T and therefore follows the desired Gaussian model. The engine applies this construction in batches.

11.3 Controlled repair

When a matrix is only slightly non-PSD, the implementation invokes a nearest-correlation-matrix repair. If the repaired matrix still cannot be factorized, it attempts a tiny $1e-12$ diagonal regularization followed by another correlation repair. If factorization remains impossible, the cell is marked invalid rather than silently accepting an unstable result.

11.4 Why silent fallback is dangerous

A covariance matrix repair changes the statistical model. A production scientific code should therefore expose the event through diagnostics. The project stores invalid covariance fraction and minimum eigenvalue, allowing downstream users to quantify how often numerical repair or rejection occurs.

11.5 Recommended audit

- Map minimum eigenvalue spatially.
- Map invalid covariance fraction by day and cell.
- Report the fraction of cells requiring repair.
- Repeat with tighter/looser PSD tolerances as a sensitivity analysis.

12. Streaming Monte Carlo Engine

The nonlinear transformation from (T, Td, P) to moisture statistics prevents exact propagation by a simple linear transformation. The engine therefore samples the joint state and applies the physical equations to every realization.

12.1 Sampling equation

$$Z \sim N(0, I_3)$$

$$X = \mu + L Z$$

$$P = \exp(X_3)$$

The first two components are temperature and dew point in °C; the third is log pressure.

12.2 Memory strategy

The full tensor with dimensions [samples, latitude, longitude] is never materialized. Instead, cells are processed in chunks and samples are processed in batches.

Parameter	Configured default	Purpose
N_SAMPLES	5000	Monte Carlo realizations per cell/day
CELL_CHUNK_SIZE	1024	Maximum spatial cells processed together
SAMPLE_BATCH_SIZE	256	Maximum Monte Carlo samples in one batch
MAX_WORKERS	2	Conservative process parallelism
RANDOM_SEED	20260821	Deterministic random stream seed

12.3 Validity filtering

A realization contributes to the transformed statistics only when all required outputs are finite and physically valid. In particular, $e \geq P$ is rejected for mixing ratio/specific humidity because the denominator $P - e$ would be non-positive.

12.4 Streaming moment accumulation

Each batch returns moment states rather than retaining its raw values. Batch states are merged across the full Monte Carlo sequence. This provides $O(\text{batch memory})$ behavior rather than $O(N_SAMPLES)$ storage.

12.5 Monte Carlo error

For a simple mean estimator, Monte Carlo standard error generally decreases approximately as $N^{-1/2}$ under independent sampling. Thus increasing N from 5,000 to 20,000 should reduce idealized mean Monte Carlo error by about a factor of two, at roughly four times the sample-generation work.

$$SE_MC(\text{mean}) \propto 1/\sqrt{N}$$

12.6 Recommended convergence experiment

N = 500, 1000, 2000, 5000, 10000

For each N:

use the same scientific configuration

repeat with independent seeds

compare mean, std, skewness, kurtosis

summarize max / median absolute change

Select N from statistical convergence, not runtime preference.

13. NetCDF Output Specification

The main output is a compressed NetCDF4 file with dimensions (doy, latitude, longitude). The diagnostic file uses the same climatological-day and spatial dimensions.

13.1 Main climatology variables

Variable	Meaning	Unit
mean_rh	Mean relative humidity	%
std_rh	Sample standard deviation of RH	%
skew_rh	Bias-corrected skewness	1
kurt_rh	Fisher excess kurtosis	1
mean_vapor_pressure	Mean vapor pressure	hPa
std_vapor_pressure	Sample standard deviation of vapor pressure	hPa
skew_vapor_pressure	Bias-corrected skewness	1
kurt_vapor_pressure	Fisher excess kurtosis	1
mean_mixing_ratio	Mean mixing ratio	kg kg-1
std_mixing_ratio	Sample standard deviation of mixing ratio	kg kg-1
skew_mixing_ratio	Bias-corrected skewness	1
kurt_mixing_ratio	Fisher excess kurtosis	1
mean_specific_humidity	Mean specific humidity	kg kg-1
std_specific_humidity	Sample standard deviation of specific humidity	kg kg-1
skew_specific_humidity	Bias-corrected skewness	1
kurt_specific_humidity	Fisher excess kurtosis	1

13.2 Coordinate and calendar variables

Variable	Role
doy	1..366 climatological slot index
latitude	degrees_north
longitude	degrees_east
month	Calendar month label associated with slot
day	Calendar day label associated with slot
reserved_day	Marks the reserved DOY 59 convention

13.3 Diagnostic variables

Variable	Interpretation
supersaturation_fraction	Fraction of Monte Carlo realizations with raw RH > 100% before clipping.
invalid_e_over_p_fraction	Fraction of realizations with finite e and P but $e \geq P$.
invalid_covariance_fraction	Fraction/indicator associated with covariance validity failures.
min_eigenvalue	Minimum eigenvalue of the correlation/covariance representation before repair.
valid_sample_count	Number of valid Monte Carlo realizations contributing to transformed moments.
corr_T_Td	Empirical correlation between T and Td.
corr_T_logP	Empirical correlation between T and logP.
corr_Td_logP	Empirical correlation between Td and logP.
valid_observation_count	Number of paired-valid historical observations.

13.4 Missing-value convention

The writer uses -9999 as the NetCDF fill value for floating-point and integer fields where appropriate. Downstream readers should respect the variable's `_FillValue` rather than interpreting -9999 as a physical observation.

14. Checkpoints, Restartability, and Reproducibility

A major engineering feature is that the calculation can be broken into durable intermediate states. Annual checkpoints preserve statistical accumulators; daily checkpoints preserve later-stage work. This is important for multi-day or multi-year jobs where a single failure should not require recomputing the entire archive.

14.1 Checkpoint layers

Layer	Content	Purpose
Annual	n, means, M2 values, cross-products	Reusable yearly sufficient statistics
Merged	Combined multi-year second-order state	Input to daily model construction
Daily	Monte Carlo/transformed moment state and diagnostics	Restart finalization and partial recovery
Final	NetCDF climatology and diagnostics	User-facing scientific product

14.2 Integrity metadata

- Configuration hash identifies the scientific/operational parameter set.
- Script SHA-256 identifies the exact implementation content used.
- Schema version distinguishes the output/checkpoint contract.
- SHA-256 checksums are used to verify checkpoint completeness and integrity.
- UTC timestamps provide provenance for generated artifacts.

14.3 Restart principle

A restart is scientifically valid only when the checkpoint was generated under the same compatible schema and configuration contract. The changelog explicitly notes that v0.6.1 preserved the v6 checkpoint/config contract so existing 1981–2020 annual checkpoints could be reused.

14.4 What must be archived

- Exact source commit or source archive.
- Configuration values and hashes.
- Input file manifest and hashes.
- Python environment specification.
- Checkpoint manifest and checksums.
- Final NetCDF files and checksums.
- Validation logs and convergence results.

15. Validation and Scientific Quality Assurance

The project includes explicit validation for calendar mapping, online statistics, higher moments, independent physical ground truth, synthetic multivariate-normal behavior, and logging consistency. The methodology documentation additionally recommends physical bounds, convergence experiments, and checkpoint checksum validation.

15.1 Calendar validation

Leap-day tests verify the custom Feb-28/Feb-29 pooling rule and March-1 shift. These tests should remain mandatory because calendar errors can be systematic and difficult to detect from aggregate annual summaries.

15.2 Numerical statistics validation

Welford/Pébay outputs should be compared against direct NumPy/SciPy calculations on synthetic arrays. The repository includes tests for Welford and Pébay behavior.

15.3 Independent physics ground truth

The project includes a vectorized physics ground-truth test. The purpose is to compare the production transformation against a separately expressed reference calculation rather than only testing a function against itself.

15.4 Synthetic MVN validation

The multivariate-normal stage should be tested using known synthetic means/covariances. Empirical Monte Carlo estimates should converge toward the theoretical state distribution and produce plausible transformed moisture statistics.

15.5 Physical bounds

Quantity	Expected condition
RH	$0 \leq \text{RH} \leq 100$ after clipping
Vapor pressure e	$e > 0$ for valid transformed samples
Pressure P	$P > 0$ by construction from $\exp(\log P)$
Mixing ratio r	$r > 0$ for valid e and P
Specific humidity q	$0 < q < 1$ for valid positive r
Covariance eigenvalues	No hard-invalid negative eigenvalues

15.6 Convergence diagnostics

A production report should not only state `N_SAMPLES`. It should quantify how output statistics change as `N` increases. Recommended metrics include absolute change, relative change, spatial maximum, spatial median, and the fraction of cells exceeding a user-defined tolerance.

15.7 Restart equivalence

A powerful test is to run a small synthetic dataset both continuously and through an interrupted checkpoint/restart workflow. The final states should agree to the expected floating-point tolerance.

16. Sensitivity Analysis Framework

The scientific method has several choices that can materially influence the output. The project methodology explicitly recommends a sensitivity matrix rather than selecting an approach based on runtime alone.

Dimension	Baseline / alternatives	Question
Monte Carlo N	500 / 1000 / 2000 / 5000 / 10000	Is N=5000 converged for all target statistics?
Pressure model	P / logP	Does the positivity transform materially affect tails?
Saturation	Water-only / phase-aware	How much do cold-season results depend on phase?
Calendar	Pooled / alternative Feb treatment	How sensitive are transition-season climatologies?
Joint model	Gaussian / mixture / copula	How important is non-Gaussian dependence?

16.1 Recommended decision rule

Choose the final configuration using quantitative validation criteria defined before examining the final maps. For example, require Monte Carlo mean changes below a specified tolerance, stable standard deviations, and no material degradation of physical validity.

16.2 Sensitivity reporting template

```
Experiment:
  baseline:
  alternative:
  variable:
  statistic:
  N / seed:
  maximum absolute change:
  median absolute change:
  relative change:
  cells exceeding tolerance:
  interpretation:
  decision:
```

16.3 Reproducibility requirement

Every sensitivity run should carry its own configuration hash and seed. Do not overwrite baseline outputs with sensitivity outputs.

17. User Guide: From Installation to Production

17.1 Environment creation

```
conda env create -f environment.yml
conda activate humidclimatologyengine
```

The environment file uses Python 3.12 and conda-forge packages, with cdsapi installed through pip. The pyproject requires Python ≥ 3.11 , so Python 3.12 satisfies the package metadata while matching the supplied environment definition.

17.2 Configure input/output paths

The current production source contains explicit Windows paths for T2m, dew point, surface pressure, and the output directory. Before running on another machine, these paths must be changed or externalized. A production-quality deployment should prefer configuration files or environment variables rather than editing scientific source code.

17.3 Acquire data

```
python scripts/download_era5land_daily_statistics.py \
--years 1981-2020 \
--variables 2m_temperature 2m_dewpoint_temperature surface_pressure \
--statistics daily_mean \
--out-dir ./data/raw
```

17.4 Run repository tests

```
pytest
```

17.5 Execute the engine

```
python moisture_climatology.py
```

The exact production runtime depends on the input archive, grid size, storage speed, available RAM, CPU count, and whether existing checkpoints are reused. Runtime should therefore be benchmarked on a representative subset rather than estimated from the number of years alone.

17.6 Production checklist

- Inputs downloaded and audited.
- All three variables have compatible grids and dates.
- Time-zone convention is explicitly recorded.
- pytest passes.
- Mandatory internal scientific tests pass.
- Monte Carlo convergence has been checked.
- Annual checkpoints are complete and checksummed.
- Final NetCDF files pass physical and structural validation.
- Configuration and source hashes are archived.

18. Operational Guide for Large Windows Jobs

The implementation is intentionally conservative for Windows and RAM-limited environments. The configured worker count is two, with an explicit comment that increasing workers can be RAM-heavy.

18.1 Memory model

Approximate sample-memory $\propto$ cell_chunk_size $\times$ sample_batch_size $\times$ number_of_state_arrays

The exact coefficient depends on the number and dtype of temporary arrays created by NumPy. Reducing CELL_CHUNK_SIZE and SAMPLE_BATCH_SIZE is therefore the first response to memory pressure.

18.2 If RAM is insufficient

- 1 Reduce MAX_WORKERS first if multiple processes duplicate large arrays.
- 2 Reduce CELL_CHUNK_SIZE.
- 3 Reduce SAMPLE_BATCH_SIZE.
- 4 Benchmark N_SAMPLES on a representative subset before a full run.
- 5 Avoid loading unnecessary xarray variables or coordinates into memory.

18.3 If CPU utilization is low

- Check whether NetCDF I/O or disk throughput is the bottleneck.
- Increase workers only after measuring peak RAM.
- Profile a small subset before changing scientific code.

18.4 If the process stops unexpectedly

- Inspect the latest checkpoint integrity.
- Verify configuration and script hashes.
- Do not delete completed annual checkpoints unless their schema is incompatible.
- Restart using the project's checkpoint-aware workflow.

18.5 Logging

The source provides a single log_progress definition and records RAM/CPU telemetry on a best-effort basis. v0.6.1 specifically fixed a duplicate log_progress definition and standardized stage logging around phase metadata.

19. Interpretation of the Four Moisture Products

19.1 Relative humidity

RH is dimensionless in ratio form but conventionally reported as percent. It is bounded after clipping. Its mean is not equivalent to RH computed from climatological mean temperature and mean dew point because the saturation vapor pressure function is nonlinear.

$$E[RH] \neq 100 e_s(E[Td]) / e_s(E[T]) \text{ in general}$$

19.2 Vapor pressure

Vapor pressure is more directly tied to the amount of water vapor than RH because it does not divide by temperature-dependent saturation pressure. Nevertheless, it remains a nonlinear function of dew point.

19.3 Mixing ratio

Mixing ratio explicitly depends on total pressure. Two cells with identical dew point can have different r because pressure differs.

$$r = 0.622 e / (P - e)$$

19.4 Specific humidity

Specific humidity is a bounded mass fraction. It is related monotonically to mixing ratio.

$$q = r / (1 + r)$$

19.5 Why all four are useful

Variable	Primary interpretation	Key dependence
RH	Degree of saturation	T and Td
e	Water-vapor partial pressure	Td
r	Water-vapor / dry-air mass ratio	Td and P
q	Water-vapor / moist-air mass fraction	Td and P

20. Error Budget and Uncertainty Sources

The output should be interpreted as a modeled climatology with multiple uncertainty layers. The Monte Carlo procedure captures uncertainty induced by the fitted joint distribution, but it does not automatically represent all real-world uncertainty in ERA5-Land.

Source	Nature	Captured by current engine?
ERA5-Land/reanalysis uncertainty	Input-data/model uncertainty	Not explicitly
Finite historical sample	Parameter estimation uncertainty	Partially represented by fitted distribution; not a full Bayesian
Gaussian approximation	Model-form error	No; it is an assumption
Monte Carlo sampling	Numerical approximation	Yes, measurable through convergence
Thermodynamic formula choice	Physical approximation	Only through sensitivity analysis
Leap-day convention	Calendar/modeling choice	Explicitly defined
Floating-point arithmetic	Numerical error	Controlled but nonzero
PSD repair	Numerical/statistical adjustment	Diagnosed but changes the fitted matrix

20.1 Important distinction: Monte Carlo is not total uncertainty

Increasing N_SAMPLES reduces Monte Carlo integration noise, but it does not remove reanalysis uncertainty, model-form uncertainty, or parameter-estimation uncertainty. A study requiring confidence intervals for the climatological parameters may need an outer bootstrap, Bayesian posterior, block bootstrap, or another uncertainty framework.

20.2 Suggested extension

A future uncertainty module could resample historical years or days, refit the joint model, and repeat Monte Carlo propagation. Such an outer loop should preserve temporal dependence where scientifically relevant rather than blindly resampling individual days.

21. Reproducibility Protocol

- 1 Freeze the source code version and record the Git commit or archive hash.
- 2 Freeze the environment with environment.yml and, for publication, record exact package versions.
- 3 Create an input manifest containing every NetCDF file, size, date range, and SHA-256.
- 4 Record the CDS request parameters and retrieval dates.
- 5 Record the calendar convention and time-zone convention.
- 6 Record all engine configuration values and the configuration hash.
- 7 Record RANDOM_SEED and Monte Carlo sample count.
- 8 Preserve annual and daily checkpoint manifests.
- 9 Run the complete test suite.
- 10 Run convergence tests and preserve their results.
- 11 Generate final NetCDF products and diagnostics.
- 12 Compute final file checksums.
- 13 Archive this guide, the README, methodology, and a machine-readable metadata record with the results.

21.1 Minimal provenance record

```
project:
  name: HumidClimatologyEngine
  version: 0.6.1
  schema_version: 6.0

period:
  start_year: 1981
  end_year: 2020

model:
  state: [T2m_C, Td2m_C, ln_surface_pressure_hPa]
  distribution: multivariate_normal

monte_carlo:
  n_samples: 5000
  cell_chunk_size: 1024
  sample_batch_size: 256
  max_workers: 2
  random_seed: 20260821

calendar:
  slots: 366
  feb28_feb29_pool: true
  reserved_doy: 59

integrity:
  config_hash: <record>
  script_sha256: <record>
```

22. Formula Compendium

This section is intended as a quick mathematical reference for reviewers, future maintainers, and users who need to reproduce the calculations independently.

22.1 Unit transformations

$$T_C = T_K - 273.15$$

$$Td_C = Td_K - 273.15$$

$$P_{hPa} = P_{Pa}/100$$

$$L = \ln(P_{hPa})$$

22.2 Empirical moments

$$\bar{x} = (1/n) \sum x_i$$

$$M2 = \sum (x_i - \bar{x})^2$$

$$s^2 = M2/(n-1)$$

$$\text{cov}(x,y) = \sum (x_i - \bar{x})(y_i - \bar{y})/(n-1)$$

$$\rho_{xy} = \text{cov}(x,y)/(s_x s_y)$$

22.3 Welford update

$$\delta = x - \text{mean_old}$$

$$\text{mean_new} = \text{mean_old} + \delta/n_{\text{new}}$$

$$M2_{\text{new}} = M2_{\text{old}} + \delta(x - \text{mean_new})$$

22.4 Parallel merge

$$n = n1 + n2$$

$$\delta = \text{mean2} - \text{mean1}$$

$$\text{mean} = \text{mean1} + \delta(n2/n)$$

$$M2 = M2_1 + M2_2 + \delta^2(n1 n2/n)$$

$$C_{xy} = C_{xy,1} + C_{xy,2} + \delta_x \delta_y (n1 n2/n)$$

22.5 Moisture physics

$$e_{s,w}(T) = 6.112 \exp[17.67T/(T+243.5)]$$

$$e_{s,i}(T) = 6.112 \exp[22.46T/(T+272.62)]$$

$$e = e_s(Td)$$

$$RH_{\text{raw}} = 100 e/e_s(T)$$

$$RH = \text{clip}(RH_{\text{raw}}, 0, 100)$$

$$r = 0.622 e/(P-e)$$

$$q = r/(1+r)$$

22.6 Gaussian sampling

$$Z \sim N(0, I)$$

$$X = \mu + LZ, \text{ Sigma} = LL^T$$

$$P = \exp(X_3)$$

22.7 Monte Carlo summaries

$$\text{mean_hat} = (1/N) \sum Y_j$$

$$\text{variance_hat} = \sum (Y_j - \text{mean_hat})^2 / (N-1)$$

$$\text{SE_MC}(\text{mean_hat}) \approx \text{SD}(Y) / \sqrt{N}$$

22.8 Bias-corrected shape statistics

$$G1 = \sqrt{n(n-1)} / (n-2) \cdot m_3 / m_2^{3/2}$$

$$G2 = (n-1) / [(n-2)(n-3)] \cdot [(n+1)b_2 - 3(n-1)]$$

$$b_2 = n M_4 / M_2^2$$

23. Worked Numerical Example

The following small example illustrates the physical layer. It is not a replacement for the production Monte Carlo workflow; it is a transparent sanity check.

23.1 Given state

Quantity	Value
T	25 °C
Td	18 °C
P	1005 hPa

23.2 Saturation pressure

$$e_s(T) = 6.112 \exp[17.67 \times 25 / (25 + 243.5)]$$

This yields a saturation vapor pressure near 31.7 hPa under the implemented approximation.

$$e = e_s(18 \text{ °C})$$

The dew-point transformation yields a vapor pressure near 20.6 hPa.

23.3 Relative humidity

$$RH \approx 100 \times 20.6 / 31.7 \approx 65\%$$

23.4 Mixing ratio

$$r \approx 0.622 \times 20.6 / (1005 - 20.6) \approx 0.0130 \text{ kg kg}^{-1}$$

23.5 Specific humidity

$$q \approx 0.0130 / (1 + 0.0130) \approx 0.0128 \text{ kg kg}^{-1}$$

The repository test uses essentially this physical sanity pattern: T=25 °C, Td=18 °C, and P=1005 hPa, then checks RH bounds, positive vapor pressure, positive mixing ratio, and $0 < q < r$.

23.6 What this example demonstrates

- The pressure unit must match the vapor-pressure unit.
- Dew point determines actual vapor pressure in this formulation.
- RH is a ratio of two nonlinear saturation pressures.
- Mixing ratio depends explicitly on pressure.
- Specific humidity is a bounded transformation of mixing ratio.

24. Test Matrix for Release Candidates

Test class	Example	Pass criterion
Calendar	Leap/non-leap mapping	Exact expected DOY values
Physics	Known T/Td/P state	Bounds and reference values agree
Welford	Synthetic random arrays	Agreement with direct statistics within tolerance
Pébay	Known moments	Agreement with SciPy/direct calculation
MVN	Known mu/Sigma	Empirical moments converge
PSD	Valid and near-invalid matrices	Correct repair/rejection behavior
Monte Carlo	N sensitivity	Statistics stabilize within tolerance
Restart	Interrupted vs continuous	Equivalent final states
I/O	NetCDF round trip	Variables, dimensions, attributes intact
Integrity	Checkpoint hashes	All referenced files verify
Performance	Representative grid	Memory/CPU targets remain acceptable

24.1 Release gate

- All automated tests pass.
- No hidden fallback or ignored exception is present in the scientific path.
- Any covariance repair rate is quantified.
- Monte Carlo convergence is demonstrated.
- Output schema is validated.
- Input provenance is complete.
- The exact source and environment can be reconstructed.

25. Limitations, Risks, and Future Extensions

25.1 Current limitations documented by the project

- Multivariate-normal assumption.
- Finite Monte Carlo error.
- ERA5-Land/reanalysis uncertainty.
- Thermodynamic approximation choice.
- Custom leap-day convention.

25.2 Additional methodological cautions

- The model is day-of-year climatological; it does not explicitly model temporal autocorrelation between neighboring days.
- Gaussian dependence may underrepresent multimodality and tail dependence.
- The current Monte Carlo layer propagates the fitted state distribution but does not by itself provide full parameter uncertainty intervals.
- The phase rule at 0 °C is a discrete convention; mixed-phase atmospheric conditions are more complex.
- The daily-statistics product's aggregation definition and time zone are part of the scientific input definition.

25.3 High-value future extensions

- 1 Add a configuration-file-first interface for all paths and scientific parameters.
- 2 Add explicit input schema validation for dimensions, units, coordinate monotonicity, and calendar completeness.
- 3 Add bootstrap or Bayesian parameter uncertainty around the joint state model.
- 4 Add Gaussian-mixture or copula alternatives for sensitivity analysis.
- 5 Add spatially coherent uncertainty modeling if downstream applications require it.
- 6 Add automated convergence stopping based on output tolerances.
- 7 Add richer provenance manifests in JSON/YAML.
- 8 Add a formal benchmark suite for memory and throughput.

26. Reviewer and Maintainer Guide

A scientific reviewer should inspect the project in the following order.

- 1 Verify the exact input product and aggregation convention.
- 2 Verify unit conversions and the pressure log transform.
- 3 Verify the leap-day mapping.
- 4 Verify paired-valid filtering.
- 5 Reproduce mean, covariance, and correlation calculations on a small synthetic dataset.
- 6 Verify the saturation vapor pressure equations and phase rule.
- 7 Verify the mixing-ratio denominator and invalid $e \geq P$ handling.
- 8 Verify the Gaussian sampling and Cholesky transformation.
- 9 Verify Monte Carlo convergence.
- 10 Verify higher-moment definitions against SciPy conventions.
- 11 Inspect diagnostics for supersaturation and covariance problems.
- 12 Verify output metadata and integrity hashes.

26.1 Questions a reviewer should ask

- Does the daily input represent the intended physical daily statistic?
- Are all three variables aligned in time and space?
- Does the custom calendar convention match the scientific objective?
- Are covariance estimates based on the same paired observations?
- How often is covariance repair required?
- How many Monte Carlo realizations are valid?
- Are skewness and kurtosis converged, not just means?
- Does the chosen Gaussian model reproduce important dependence features?
- Can a third party reconstruct the exact run from the archived metadata?

26.2 Maintainer rule

Any change to the scientific formulas, calendar mapping, state representation, validity mask, moment definitions, random-number generation, or NetCDF schema should be treated as a scientific-version change. Do not hide such changes inside a patch release without documenting their impact.

27. References and Further Reading

The following references support the scientific foundations or data provenance of the implementation. The first group contains official data documentation; the second group contains statistical and thermodynamic foundations.

27.1 ERA5-Land and Copernicus

Copernicus Climate Change Service (C3S). ERA5-Land post-processed daily statistics from 1950 to present. Dataset documentation and metadata. DOI: 10.24381/cds.e9c9c792. The official catalogue describes daily statistics derived from ERA5-Land hourly data, including daily mean/min/max options, sub-daily frequency, and time-zone selection.

Copernicus Climate Change Service (C3S). ERA5-Land hourly data from 1950 to present. DOI: 10.24381/cds.e2161bac. Official dataset documentation and provenance.

27.2 Online/mergeable statistics

Welford, B. P. (1962). Note on a Method for Calculating Corrected Sums of Squares and Products. *Technometrics*, 4(3), 419–420. DOI: 10.1080/00401706.1962.10490022.

Pébay, P. (2008). Formulas for Robust, One-Pass Parallel Computation of Covariances and Arbitrary-Order Statistical Moments. Sandia technical report. This family of formulas underpins the mergeable higher-moment strategy used by the project.

27.3 Thermodynamic humidity relationships

Sonntag, D. (1990). Important reference for Magnus-type saturation vapor pressure formulations over water and ice. The supplied scientific methodology uses a phase-aware Magnus-style approximation; coefficient conventions should always be reported exactly because several variants exist.

Smithsonian Meteorological Tables / standard meteorological definitions. Atmospheric humidity definitions for vapor pressure, saturation vapor pressure, mixing ratio, and specific humidity. A publicly accessible copy is hosted through Unidata.

27.4 Software-specific references

HumidClimatologyEngine README.md. Project scientific rationale, formulas, workflow, validation, limitations, and sensitivity matrix.

HumidClimatologyEngine docs/methodology.md. Condensed implementation methodology.

HumidClimatologyEngine src/moisture_climatology_v6.py. Production implementation inspected for this guide.

27.5 Useful official links

ERA5-Land daily statistics: <https://cds.climate.copernicus.eu/datasets/derived-era5-land-daily-statistics>

ERA5-Land hourly data: <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land>

Welford DOI: <https://doi.org/10.1080/00401706.1962.10490022>

Appendix A. Configuration Reference

Parameter	Configured value	Scientific/operational role
START_YEAR	1981	First climatological year
END_YEAR	2020	Last climatological year
DOY_COUNT	366	Number of climatological slots
N_SAMPLES	5000	Monte Carlo sample count
CELL_CHUNK_SIZE	1024	Spatial batching
SAMPLE_BATCH_SIZE	256	Sample batching
MAX_WORKERS	2	Conservative parallelism
RANDOM_SEED	20260821	Reproducible random sequence
PSD_TOL	1e-10	PSD numerical tolerance
PSD_REPAIR_TOL	1e-8	Maximum repairable negative eigenvalue magnitude
MIN_OBS	3	Minimum configured historical observation threshold
MIN_MC_VALID	10	Minimum valid Monte Carlo sample threshold
SCHEMA_VERSION	6.0	Output/checkpoint schema
CHECKPOINT_VERSION	6.0	Checkpoint compatibility version

A.1 Important path settings

Setting	Current source value
T2M_DIR	F:\Kazemi\era5\land\daily\T2m
D2M_DIR	F:\Kazemi\era5\land\daily\Dew_Point_Temperature
SP_DIR	F:\Kazemi\era5\land\daily\Surface_Pressure
OUTPUT_DIR	C:\c

These paths are machine-specific and should be replaced for another environment. They are documented here because they are part of the inspected implementation, not because they are universally recommended.

Appendix B. Practical Troubleshooting

B.1 Import or environment error

Verify the active environment, Python version, netCDF4 installation, and xarray compatibility. Run `pytest` before starting a large production job.

B.2 Missing input file

Check the expected year/month directory, file naming convention, and retrieval manifest. Do not substitute a file from another daily aggregation convention without recording the change.

B.3 Coordinate mismatch

Temperature, dew point, and pressure must refer to compatible grids. A silent grid mismatch can produce scientifically invalid covariance.

B.4 Too many invalid Monte Carlo samples

- Inspect `invalid_e_over_p_fraction`.
- Inspect the minimum eigenvalue and invalid covariance diagnostics.
- Verify pressure units are hPa before moisture equations.
- Verify the log-pressure model is reconstructed with `exp()`.
- Inspect extreme sampled temperatures/dew points.

B.5 Non-converged skewness/kurtosis

Increase `N_SAMPLES`, repeat with independent seeds, and examine the stability of the transformed tails. Do not assume that a stable mean guarantees stable higher moments.

B.6 High RAM usage

Lower `MAX_WORKERS`, `CELL_CHUNK_SIZE`, or `SAMPLE_BATCH_SIZE`. The safest order is to reduce process parallelism first because each process can carry substantial temporary arrays.

B.7 Restart rejected

Compare schema version, checkpoint version, configuration hash, and script hash. If a scientific parameter changed, treat the old checkpoint as belonging to a different experiment unless compatibility is explicitly established.

Appendix C. Publication-Ready Reporting Template

A paper, technical report, or data release using HumidClimatologyEngine should report at least the following information.

- ERA5-Land product name, DOI, retrieval date, and aggregation definition.
- Spatial grid and temporal coverage.
- Variables and source units.
- Unit conversion equations.
- Calendar convention, including leap-day treatment.
- Paired-valid criterion.
- State vector $X=(T,T_d,\ln P)$.
- Covariance estimation method.
- Multivariate-normal assumption.
- PSD/Cholesky treatment and repair thresholds.
- Saturation vapor pressure equations and phase rule.
- RH, e, r, and q equations.
- Monte Carlo sample count, batch size, cell chunk size, and random seed.
- Moment estimators and bias corrections.
- Physical and numerical validation tests.
- Monte Carlo convergence results.
- Checkpoint/integrity strategy.
- Software version, source commit, and environment.

C.1 Suggested methodological sentence

“Daily moisture climatologies were generated by fitting a paired-valid trivariate normal model to 2-m air temperature, 2-m dew-point temperature, and log surface pressure for each climatological day and grid cell; Monte Carlo realizations were transformed through phase-aware saturation vapor pressure, relative humidity, mixing-ratio, and specific-humidity equations, and transformed moments were accumulated with streaming mergeable estimators.”

C.2 Reproducibility statement

“All input manifests, configuration parameters, software hashes, checkpoint checksums, convergence diagnostics, and NetCDF output hashes were archived with the analysis.”

28. Final Scientific Assessment

The supplied HumidClimatologyEngine is more than a humidity calculator: it is a reproducible statistical-physical pipeline that explicitly separates data acquisition, joint-state estimation, nonlinear thermodynamics, Monte Carlo propagation, numerical validation, checkpointing, and final serialization.

Its strongest methodological feature is the decision to preserve the joint structure of temperature, dew point, and pressure before deriving humidity variables. Its strongest engineering feature is the combination of mergeable statistics, streaming Monte Carlo, durable checkpoints, and explicit integrity metadata.

The most important scientific caution is equally clear: the outputs inherit the limitations of ERA5-Land, the Gaussian joint-distribution approximation, the chosen saturation vapor pressure formulation, the custom calendar convention, and finite Monte Carlo sampling. These are not defects to conceal; they are model assumptions that should be quantified through sensitivity and convergence experiments.

For a publication-grade workflow, the next level of rigor is not simply adding more code. It is demonstrating convergence, documenting the exact input aggregation convention, quantifying covariance repair, validating the nonlinear physical layer independently, and preserving complete provenance so another researcher can reconstruct the result.

End of reference guide.

Prepared from the supplied HumidClimatologyEngine archive, with external scientific/data references cross-checked against official and primary sources.